Michelle Elias

M.Sc. Biophysics Student · Computational Systems Medicine

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König Lab profile
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Biophysics master's student in the König Lab (Systems Medicine of the Liver). Background in PBPK/PD modelling, digital twins of drugs and reproducibility studies of published models. Master project: AI-based segmentation of liver histology feeding into image-informed PDE–ODE simulations.

EDUCATION

M.Sc. Biophysics

10/2025 – expected 10/2027

Humboldt-Universität zu Berlin — Berlin

- **Master project & thesis** (working title): *From spatial liver imaging to image-informed dynamical simulation*
  Project 08/2026 – 01/2027 (HU Berlin), thesis 02/2027 – 09/2027 (Universität zu Lübeck); supervisor Prof. Dr. Matthias König
  Reproducible workflow from multimodal whole-slide liver histology to AI-assisted segmentation, quantitative feature maps and image-informed PDE–ODE simulation (FEniCSx, libRoadRunner); use cases: APAP-induced necrosis and hepatic steatosis. slideviz
- **Focus: Systems Biology** — Dynamic Modelling and Data Integration (**1.3**); Computational Analysis and Interpretation of High-throughput Data (**1.3**), incl. project on CNN-based prediction of ribosome load from 5′UTR sequences (PyTorch, Captum). Code

B.Sc. Biology

Final grade 1.9

10/2019 – 04/2025

Humboldt-Universität zu Berlin — Berlin

- **Bachelor project (1.0) & thesis (1.0):** *A physiologically based pharmacokinetic (PBPK) model of the sulfonylurea glimepiride* — Systems Medicine of the Liver (König Lab). Whole-body PBPK model covering renal and hepatic impairment, bodyweight and CYP2C9 genotype; validated against 20 clinical studies and published as a digital twin [5].

Abitur

2013 – 2019

Georg-Büchner-Gymnasium — Berlin

Bilingual school (German/English); advanced courses in Mathematics and Visual Arts.

RESEARCH EXPERIENCE

Student Assistant, DFG Priority Programme SPP 2311

08/2026 – present

University of Stuttgart, Institute for Statics and Dynamics of Aerospace Structures (ISD) · with the König Lab

- Developing **VisFEM**, a web platform for interactive visualization of finite-element (FEM) simulation results (SPP 2311 start-up funding project); first-author manuscript in preparation [13].

Student Assistant, Systems Medicine of the Liver (König Lab)

01/2026 – 07/2026

Humboldt-Universität zu Berlin, Institute for Theoretical Biology — Berlin

- PBPK/PD modelling and digital twins; built reproducible model pipelines for Physiome reproducibility studies (dapagliflozin, losartan, hydrochlorothiazide, canagliflozin, empagliflozin, rivaroxaban).
- Contributed to VisFEM for SPP 2311; coordinated programme-wide data collection by designing and distributing project and model surveys to all subprojects.
- Supported manuscript development and editing for ongoing group publications.

Research Intern, Cardiovascular Precision Medicines 04/2024 – 06/2024
Bayer AG, Pharmaceuticals R&D Wuppertal

- Cultured hiPSC-derived cardiomyocytes; gene knockdown via siRNA transfection.
- qPCR, Western blot, DNA/RNA/protein extraction; statistical analysis and visualization of results. Excellent internship reference.

AWARDS

10/2026 **Open Science Award 2026, University of Lübeck** (team award, €1,000) for *An open, reproducible digital twin of hydrochlorothiazide pharmacokinetics and pharmacodynamics*, with A. Schwaiger, M. Myshkina and M. König. Contribution: reproducibility study [9].

PRESENTATIONS

05/2026 **VisFEM — interactive visualization of FEM simulation results.** Project presentation and demo, SPP 2311 Annual Meeting, BioCity Leipzig, Germany.

PUBLICATIONS

JOURNAL ARTICLES

[1] **Reproducibility of a Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin**
Elias M, Myshkina M, Nemitz N, König M. *Physiome, 2026.* **Published**
doi:10.36903/physiome.31368556 | Code

[2] **Reproducibility of a Digital Twin of the Angiotensin II Receptor Blocker Losartan**
Myshkina M, **Elias M**, Tensil E, König M. *Physiome, 2026.* **Published**
doi:10.36903/physiome.31359823 | Code

[3] **A Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin in Type 2 Diabetes Mellitus: The Effect of Dosing, Hepatorenal Impairment, and Food**
Nemitz N, **Elias M**, König M. *Pharmaceutics, 2026.* **Published**
doi:10.3390/pharmaceutics18030287 | Preprint | Code

[4] **Reproducibility of a Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**
Elias M, König M. *Physiome, 2025.* **Published**
doi:10.36903/physiome.28379193 | Code

[5] **A Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**
Elias M, König M. *Frontiers in Pharmacology, 2025.* **Published**
doi:10.3389/fphar.2025.1686415 | Preprint | Code | Web app

PREPRINTS

[6] **An Open and Reproducible Digital Twin for Personalised Anticoagulant Therapy: Modelling Apixaban Pharmacokinetics and Pharmacodynamics**
Myshkina M, Babaeva M, Nguyen LC, Bedón S, Metternich J, **Elias M**, König M. *Preprints, 2026.* **Preprint**
doi:10.20944/preprints202605.0249.v1

[7] **A Digital Twin of Empagliflozin Pharmacokinetics and Pharmacodynamics**
Alejandro J, **Elias M**, Babaeva M, König M. *Preprints, 2026.* **Preprint**
doi:10.20944/preprints202603.1559.v1 | Code

[8] **A Digital Twin of Canagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Tereshchuk V, **Elias M**, König M. *Preprints, 2026.* **Preprint**
doi:10.20944/preprints202601.2095.v1 | Code

SUBMITTED MANUSCRIPTS

[9] **Reproducibility of a Digital Twin of Hydrochlorothiazide Pharmacokinetics and Pharmacodynamics**
Elias M, Myshkina M, Schwaiger A, König M. *Physiome.* **Submitted**

[10] **Reproducibility of a Digital Twin of Canagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Elias M, Myshkina M, Tereshchuk V, König M. *Physiome.* **Submitted**
Code

[11] **Reproducibility of a Digital Twin of Empagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Elias M, Myshkina M, Alejandro J, Babaeva M, König M. *Physiome.* **Submitted**
Code

[12] **Rivaroxaban Pharmacokinetics and Coagulation Effects: Reproducibility of a Physiologically Based Modeling Approach**
Myshkina M, **Elias M**, Casabianca E, König M. *Physiome.* **Submitted**
Code

IN PREPARATION

[13] **VisFEM: A Web-Based Platform for FAIR Visualization and Sharing of Finite Element Simulations in Biomedical Research**
Elias M et al. **In preparation**
Code

SKILLS

Modelling	PBPK/PD modelling, digital twins, ODE & multiscale simulation, population simulation, parameter estimation, pharmacogenomics (CYP2C9)
Data science & AI	Machine learning (PyTorch, Captum), CNN sequence models, biomedical image analysis & segmentation, statistics & hypothesis testing, data visualization
Programming	**Python** (NumPy, SciPy, pandas, matplotlib, seaborn), R, Linux, HTML/CSS; SBML/libSBML, sbmlsim, sbmlutils, libRoadRunner
Open science	Reproducible workflows, FAIR data, SBML/OMEX/COMBINE standards, clinical data curation (PK-DB), systematic literature review
Tools	git/GitHub, Docker, VS Code, PyCharm, Jupyter, Marimo, LaTeX, Typst, BioRender
Laboratory	Cell culture (hiPSC-derived cardiomyocytes), siRNA transfection, PCR/qPCR, gel electrophoresis, Western blot
Languages	German (native) · English (C1, TOEFL iBT 113) · Polish (native comprehension, basic conversation)
Certificates	Data Analyst with Python · Microsoft Excel

FURTHER EXPERIENCE

04/2025 – 12/2025	**Student Assistant, Medical Controlling** · Martin Luther Krankenhaus, Berlin — digitization of patient records; onboarding of new assistants.
09/2024 – 12/2025	**Student Assistant, Hospital Archive** · Ev. Elisabeth Klinik, Berlin — co-coordinated the archive relocation; digitization and cataloguing of records.
01/2023 – 12/2023	**Student Assistant, Hospital Administration** · Martin Luther Krankenhaus, Berlin — patient records and insurance processing. Excellent reference.
2022	**Student Assistant, Ukrainian Refugee Center** · Johanniter, Berlin — registration, orientation and referral of refugees.
2021	**Student Assistant, COVID-19 Vaccination Center** · Malteser, Berlin — support of medical staff and patient flow.